

Mohanad Ahmed Shehata

Software Engineer | Full Stack Developer

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PROFESSIONAL SUMMARY

Software Engineer driven by precision, performance, and scalable system design. Ranked among the top of his class with a 3.9/4.0 GPA and awarded 1st Place Academic Excellence (Jan 2024, Electronics & Communications Engineering, IAET – Egyptian Aviation Academy).

With over two years of hands-on freelance experience, builds production-ready applications using React and Laravel, applying solid engineering principles rather than just writing code. Strong focus on clean architecture, separation of concerns, and design patterns to deliver systems that are maintainable, extensible, and efficient under real-world load.

Engineers robust solutions end-to-end — from structured API layers and role-based access control to real-time event-driven systems and full-text search — while consistently optimizing for performance, scalability, and reliability.

SKILLS

Languages: C++, Python, Java, JavaScript (ES6+), TypeScript, PHP

Engineering: Data Structures & Algorithms, OOP, System Design, Clean Architecture, Design Patterns

Backend: Laravel, PHP, RESTful APIs, WebSockets (Pusher / Laravel Echo), Queue & Job Processing

Frontend: React.js, HTML5, CSS3, Bootstrap

Databases: MySQL, SQL Server, Redis, Meilisearch

DevOps & Tools: Docker, CI/CD, Git, GitHub, DNS, Web Security, Postman, SEO Optimization

Soft Skills: Problem-solving, Analytical thinking, Cross-functional collaboration, Attention to detail, Adaptability

WORK HISTORY

Full Stack Software Engineer

Jun 2025 – Present

Freelance | Remote, Egypt

- **Architected and delivered** full-stack client applications, evaluating technology stacks based on scalability, security, and long-term maintenance trade-offs.
- **Eliminated critical failure paths** through systematic unit and load testing, achieving a 70% reduction in system failure rate on high-traffic customer-facing services.
- **Established reusable component and service patterns** with Laravel and React — accelerating delivery cadence across successive client engagements.

Full Stack Software Engineer

Sep 2024 – Present

Freelance | Remote, Egypt

- **Designed layered application architectures** using Repository & Services patterns within MVC, separating business logic, data access, and presentation layers for maintainability.
- **Increased user engagement by 30%** by implementing a real-time push-notification system via WebSockets (Pusher), replacing latency-prone polling across multiple platforms.
- **Reduced search latency to sub-50ms** by integrating Meilisearch via a Scout driver abstraction, replacing SQL LIKE queries across datasets of thousands of records.

- **Improved application throughput** by routing expensive operations through Redis-backed job queues, decoupling heavy computations from the HTTP request cycle.
- **Hardened security posture** by implementing OTP authentication, mail verification, and role-based access control (guards, gates, policies) across all user and admin surfaces.

PROJECTS

NEWMO — News Portal Platform

Jun – Aug 2025

PHP / Laravel | MySQL | Redis | Meilisearch | WebSockets

- Engineered a multi-role publishing system with strict domain boundaries between editorial, author, and reader tiers, enforced via a policy-gate authorization model.
- Offloaded cache invalidation, email dispatch, and notification fanout to a Redis job queue — preserving consistent HTTP response times under write-heavy editorial workflows.
- Delivered an SEO-optimised, server-rendered interface with structured metadata and real-time breaking-news push notifications.

ParkingMO — Parking Management System

Jan 2025

PHP / Laravel | MySQL | Redis | WebSockets

- Modelled the domain across five entities — customers, vehicles, slots, subscriptions, and billing cycles — with explicit state machines governing occupancy and subscription lifecycle.
- Automated the billing engine to compute accrued dues on subscription termination, eliminating manual calculation errors regardless of plan duration or slot type.
- Delivered a role-scoped admin analytics dashboard covering revenue, throughput, and utilisation, powered by indexed aggregation queries.

CERTIFICATIONS

- **React Frontend Web Developer** — Digital Egypt Pioneers Program (DEPI) / Ministry of Communications & IT | *Dec 2025*
- **Full Stack Web Development Diploma (200 hrs)** — WE | *Nov 2024*
- **Mastering OOP Using C++** — Mahara-Tech / ITI Platform | *Apr 2026*
- **Problem Solving (Basic)** — HackerRank | *Apr 2026*
- **SQL Basic & Intermediate** — HackerRank | *Mar 2026*
- **Introduction to SQL** — DataCamp | *Feb 2026*
- **Business English Track** — DEPI / Ministry of Communications & IT | *Sept 2025*

EDUCATION

B.Sc. Communication & Electronics Engineering

Expected 2027

Institute of Aviation Engineering and Technology (IAET) / Egyptian Aviation Academy | Giza, Egypt

GPA: 3.9 / 4.0 | 1st Place — Academic Excellence Award, January 2024

LEADERSHIP & ACTIVITIES

- **IEEE IAET Technical Committee — Member, 2024.** Recognised for two technical contributions in engineering workshops and cross-chapter knowledge-sharing events.
- **Digitopia Competition (Software & AI) — Stage 1 Completion, Jan 2025.** National competition by Egypt's Ministry of Communications & Information Technology.
- **Career Directions Program — EFE-Egypt, Sept 2024.** Completed professional development training in workplace communication and career readiness.